K L UNIVERSITY

Department of Electronics and Communication Engineering

CAPSTONE PROJECT REVIEW PROGRESS RECORD

Review Stage: Review–1	**Date of Review: 03 / 10 / 2026**
Project Title:	Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators
Cohort / Batch:	2024 – 2028 (Batch B4)
Project Guide:	Dr. Arun Nimmala, Associate Professor, Dept. of ECE

S. No.	Student Name	Student ID	Signature
1	**Velivala Keerthi**	2320040083	
2	**T. Venkata Ram Charan Reddy**	2320040017	
3	**Shaik Hanifa**	2320040009	
4	**K. Dinesh**	2320040095	

Brief Description of the Project

Briefly explain the premise of the project, the problem being addressed and the proposed approach. (less than 200 words)

Deep Convolutional Neural Network (CNN) inference on resource-constrained edge hardware is dominated by Multiply-Accumulate (MAC) units, where 8×8 multipliers consume substantial silicon area and power (68 LUTs on Xilinx Zynq-7020). Approximate computing offers significant energy and area reduction by relaxing arithmetic precision. However, uniform approximation across an entire network causes severe accuracy degradation or yields negligible hardware savings. Existing layer-wise mapping techniques require multi-day genetic search (e.g., ALWANN requires 7.5 CPU-days) or costly GPU retraining loops. This project develops a lightweight, single-pass sensitivity scoring metric computed in just 0.2 seconds without retraining or iterative search. By coupling static weight variance with dynamic first-order loss gradients extracted across 256 calibration images, our framework accurately quantifies layer-wise error resilience. Error-critical early layers (conv1) are allocated exact multipliers, intermediate layers receive compensated designs, and noise-resilient deep layers (conv4, accounting for 48.5% of total MACs) are assigned aggressive truncated multipliers. The heterogeneous datapath is realized as a synthesizable AXI4-Stream accelerator IP core for physical deployment on the Xilinx Zynq-7020 (PYNQ-Z2) FPGA SoC, maximizing silicon efficiency while preserving baseline classification accuracy.

Progress Achieved up to This Review

Briefly mention the work completed, implementation/simulation/experimental progress, results obtained and other major milestones achieved up to this review.

1. **RTL Multiplier Library Design & Characterization:** Designed synthesizable Verilog modules for four 8×8 unsigned multiplier architectures (Exact, Truncated-4, Compensated-4, and OR-based) using explicit partial-product arrays. Conducted exhaustive behavioral simulation of all 65,536 input vectors in Icarus Verilog, extracting Mean Error (ME), MAE, NMED, and maximum error bounds.
2. **Circuit-Level Dominance Proof:** Synthesized the multiplier library in Xilinx Vivado targeting the Zynq-7020 FPGA. Proved that Compensated multipliers strictly dominate Truncated designs—both consume identical silicon resources (56 LUTs, –17.6% area vs. 68 LUTs exact) because the synthesis engine folds the +2^(T-1) (+8) rounding constant into existing carry-in logic at zero extra LUT cost, while cutting arithmetic error by ~35% (ME = –4.25 vs –12.25).
3. **Baseline CNN Architecture & INT8 Quantization:** Trained a 4-convolutional layer CNN on Fashion-MNIST in TensorFlow/Keras, establishing an exact 32-bit floating-point baseline test accuracy of 84.69%. Designed an affine INT8 quantization pipeline featuring unsigned 8-bit activations [0, 255] and signed-magnitude weights matching the unsigned hardware multiplier datapath.
4. **Sensitivity Metric Formulation (S1–S5):** Formulated candidate sensitivity metrics. Implemented a single-pass automatic differentiation probe using TensorFlow GradientTape across 256 calibration images (0.2s runtime). Revealed that conv1 is overwhelmingly sensitive (S=1.000) due to primitive edge extraction, whereas conv4 is resilient (S=0.014).
5. **Preliminary Sweep & Noise Floor Diagnosis:** Executed the 12-cell layer-by-layer validation sweep using TRUNC=4. Discovered that the CNN's inherent resilience absorbed the mild TRUNC=4 noise (NMED = 0.000188), producing accuracy drops within ±0.1% to 0.3% (statistical noise floor) and yielding an initial NO-GO correlation ($|\rho| < 0.40$). Formulated the plan to scale to TRUNC=6 for signal amplification.

Signature of Project Guide: _____ **Date: 03 / 10 / 2026**

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CAPSTONE PROJECT REVIEW PROGRESS RECORD

Review Stage: Review–2	**Date of Review: 20 / 10 / 2026**
Project Title:	Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators
Cohort / Batch:	2024 – 2028 (Batch B4)
Project Guide:	Dr. Arun Nimmala, Associate Professor, Dept. of ECE

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Progress Achieved up to This Review

Briefly mention the work completed, implementation/simulation/experimental progress, results obtained and other major milestones achieved up to this review.

1. Noise-Floor Diagnosis & Signal Amplification (TRUNC=6): Mathematically analyzed the TRUNC=4 sweep failure, identifying that testing against the binomial noise floor invalidated ranking correlation. Scaled the approximate multiplier library to TRUNC=6 (discarding columns 0 to 5; 21 of 64 partial products), amplifying NMED by 6.6× (0.001234) and achieving 32.5% technology-independent gate reduction (247 gates vs 366 exact).

2. Formal Research Gate Clearance (GO Decision): Re-ran the 12-cell layer-wise sweep with TRUNC=6. A clear damage contrast emerged: conv1 suffered a heavy 1.85% drop while conv4 dropped only 0.41%. Our proposed combined metric (S3) achieved a Spearman rank correlation of $|\rho| = 0.80$, comfortably exceeding the pre-agreed $|\rho| \geq 0.70$ threshold to officially clear the research GO gate.

3. The conv4 Computational Asymmetry Discovery: Calculated network-wide computational density, proving that conv4 executes 1,806,336 MACs—representing 48.5% of all multiplications in the entire network—yet possesses near-zero sensitivity (S=0.014). Assigned Truncated-6 to conv4, enabling 32.5% area reduction across half the entire network's compute engine while preserving 84%+ baseline accuracy.

4. NSGA-II Multi-Objective Optimization Baseline: Implemented an evolutionary multi-objective search algorithm (DEAP library) across all 256 layer-multiplier combinations (4 layers × 4 multipliers), optimizing Accuracy Loss vs. MAC-Weighted LUT Cost. Generated the ground-truth Pareto front.

5. 3,000,000× Search Speedup Demonstration: Proved that our proposed single-pass sensitivity assignment (conv1=Exact, conv2/3=Comp-6, conv4=Trunc-6) lands directly at the Pareto-optimal knee (84.02% accuracy, 57.2 LUTs) in 0.2 seconds. Benchmarked against ALWANN (7.5 CPU-days, 10,800 min) and localized NSGA-II (~30 min), demonstrating a >3,000,000× reduction in search latency with zero Pareto quality loss.

Signature of Project Guide: _____ **Date: 20 / 10 / 2026**

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CAPSTONE PROJECT REVIEW PROGRESS RECORD

Review Stage: Review–3	**Date of Review: 06 / 11 / 2026**
Project Title:	Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators
Cohort / Batch:	2024 – 2028 (Batch B4)
Project Guide:	Dr. Arun Nimmala, Associate Professor, Dept. of ECE

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Brief Description of the Project

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Progress Achieved up to This Review

Briefly mention the work completed, implementation/simulation/experimental progress, results obtained and other major milestones achieved up to this review.

1. **Complete Synthesizable Accelerator RTL Microarchitecture:** Engineered a fully parameterizable CNN accelerator IP in synthesizable Verilog (`cnn_accelerator_top.v`). Implemented a 2-bit dynamic mode controller (`layer_controller.v`), a signed MAC unit with sign-magnitude math (`mac_unit.v`), a 3×3×Cin Processing Element with finite-state control (`conv_pe.v`), and the unified multi-mode multiplier (`axm_8x8.v`).
2. **Exhaustive 262,144-Vector RTL Verification (100% Pass):** Developed an automated verification suite (`verify_rtl_logic.py`) exhaustively simulating all 4 multiplier modes across all 65,536 input combinations. Verified 262,144/262,144 vectors with zero functional mismatches against golden mathematical models.
3. **Vivado SoC Integration & Bitstream Automation:** Authored complete Vivado automation scripts (`vivado_synth.tcl` and `vivado_block_design.tcl`). Partitioned the system into Processing System (dual ARM Cortex-A9) and Programmable Logic (accelerator IP clocked at 100 MHz), interfaced via AXI DMA using high-performance HP0 ports for streaming 17-bit packed beats ({w_sign, w_mag, act_in}).
4. **PYNQ-Z2 On-Board Deployment & Jupyter Driver:** Developed the physical hardware driver (`pynq_inference.py`) and an interactive 7-cell Jupyter notebook (`pynq_cnn_demo.ipynb`). Pre-extracted INT8 quantized weights (`weights_int8.npy`) for direct DDR3 memory DMA transfers and real-time on-board latency/FPS measurement.
5. **Publication-Ready Deliverables & GitHub Repository:** Generated all 7 publication-grade figures (`fig1`–`fig7`) for the IEEE manuscript, authored a complete 20-slide technical presentation with embedded figures and full speaker notes, and established a fully documented, open-source GitHub repository with MIT License.

Signature of Project Guide: _____ **Date: 06 / 11 / 2026**